An Indicator and Min-Cost Approach for Shoreline Extraction From Satellite Imagery in Muddy Coasts

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***Abstract*— Shoreline location plays a key role in coastal research, management, and engineering. Remote sensing enables the quantification of shoreline information with the large spatial extent and high temporal frequency. Driven by river discharge and ocean dynamics, muddy coasts exhibit complicated spatiotemporal variations. It is essential, yet challenging to extract effective shoreline features from satellite images. Taking the Yellow River Delta coast in China as the study area, we present an indicator and min-cost approach to extract the shoreline in muddy coasts. The shoreline is represented as a set of linearly connected central points with high shoreline probabilities, and a set of image and spatial indicators are developed to assess these probabilities. The Salient Value indicator integrates the gradient magnitude and the edge intensity to detect the boundary strength; the Regional Difference indicator separates the water/land class from edge intensity to measure the possibility of being water or land; and the Seaward Distance indicator spatially distinguishes the true shoreline from other spectrally similar boundaries. A cost function combines these indicators to evaluate the local shoreline possibilities. A shoreline set is produced by an improved min-cost path method to evaluate the overall shoreline possibilities. The optimal shoreline paths are selected based on the parameter analyses of the shoreline set. The performance of the approach is confirmed by comparing with the ground truth and state-of-the-art methods. The effectiveness of the approach is tested for different spatial resolution data and coastal environments.**

***Index Terms*— Geospatial information, muddy coast, satellite image, shoreline extraction, the Yellow River Delta (YRD).**

I. Introduction

SHORELINE information plays a key role in understanding the response of the coastal system to climatic changes and human activities, and it is also crucial for coastal management and engineering applications [1]. Muddy coast is one of the primary coastal types and extensively distributed around river deltas in the world [2]. Driven by river discharge and ocean dynamics, muddy coasts are highly dynamic. Obtaining muddy shoreline positions is difficult by means of *in situ* surveys, whereas remote sensing has the advantages of large areal coverage, short-term operation, and fitness for muddy shoreline extraction [3], [4]. The primary data sources for shoreline determination include a variety of aerial and satellite data, in which Moderate Resolution Imaging Spectroradiometer (MODIS) and Landsat are available in large amounts, free for operational use, and suitable for large-scale applications [5]. In river deltas that experience dramatic shoreline changes, Landsat data can capture a large proportion of spatial characteristics [6]. MODIS data have the obvious advantage in capturing rapid dynamics and have been successfully applied to extract various land–water boundaries [7]–[9].

There are numerous methods that can be used for shoreline extraction from remotely sensed data. For pixel-level methods, shoreline detection can be reduced to a classification problem. There are two classification schemes in practice. One scheme labels a pixel as edge or non-edge [10]. These edge pixels can be developed into boundaries that separate different imaging areas [11]. The other scheme is image segmentation, which classifies a pixel into land or sea type. The boundary between the land and sea pixel areas can be viewed as the shoreline. In muddy coasts, it is hard to characterize a shoreline continuously and completely by edge detection [12], whereas image segmentation suffers from more post-processing work and difficulties, such as determining reliable thresholds or homogeneous criteria in complicated coastal environments [13]. Image signals of muddy coasts always lack sufficient discrimination between land and water areas [14]; meanwhile, many other kinds of spectrally similar boundaries spread along the muddy coast, which are hard to be distinguished from the shoreline only by image features. Therefore, high-level image and spatial information are needed for muddy shoreline detection. Integration of such information may produce better results than those obtained by a single method [15].

For sub-pixel measurements, current approaches for shoreline extraction can also be classified into two schemes. One is based on soft classification and pixel unmixing techniques, in which the preliminary step is to locate mixed pixels containing shoreline information [5], [16]. This is not a trivial matter in muddy coasts, due to the complex spectral variations caused by fast water-sediment movement [17]. The other is based on sub-pixel edge localization techniques [18]. The most accurate edge location can be obtained when the edge is as close to the center of the data segment or analytic window as possible [19], [20]. The achieved accuracy of edge location

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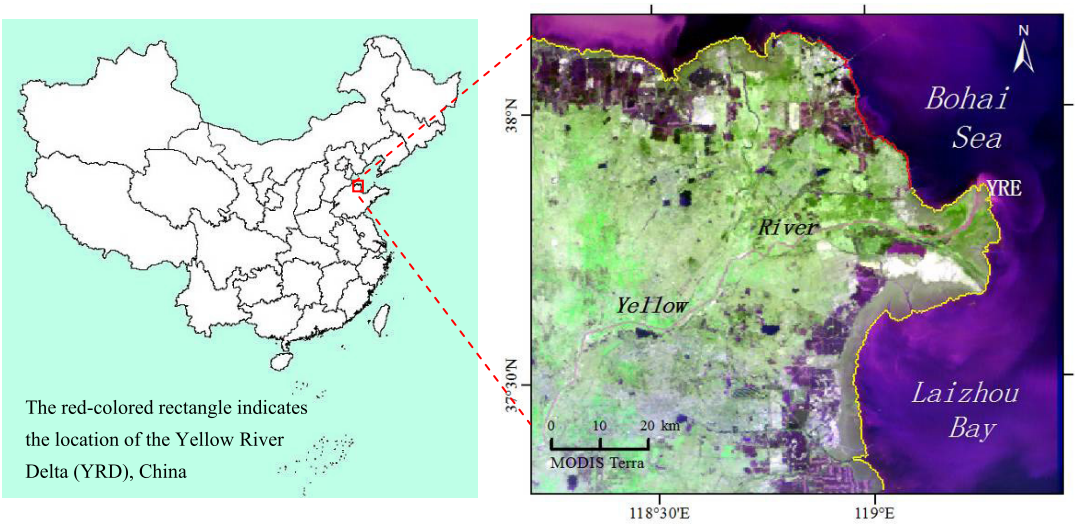


Fig. 1. Location of the study area with an RGB composite map derived from MODIS data, where the yellow and red lines are generated by the proposed approach, indicating the muddy and artificial shorelines, respectively.

decreases with increasing blurring of the boundaries [21], as is often the case in muddy coasts. The above research findings show that accurate determination of shoreline locations at pixel level is also essential for sub-pixel shoreline extraction in muddy coasts.

Up to now, many shoreline extraction methods have been applied for artificial, sandy, rocky, and glacier coasts [9], [22]–[24], but there has been little attention paid to muddy coasts. One of the reasons is the challenges come from the image and spatiotemporal characteristics of muddy coasts we mentioned above. In this article, we present an indicator and a min-cost approach for extracting muddy shorelines, which is computationally simple and easy to integrate low- and high-level cues. This article aims at performing convenient determination of the muddy shoreline with accurate geographical location, reliable geometric shape, and little human intervention.

II. Related Work

Shoreline extraction from remotely sensed data originates from cartographic mapping in which human interpretative skills are applied to compile a map from an aerial photograph. This process is characterized by the work of feature delineation, identification, and attribution [25]. In recent decades, automatic methods are developed to reduce the burden of manual tracing of shoreline features. Lee and Jurkevich [26] used a set of image-processing and edge tracing algorithms to detect the muddy shorelines in the Chesapeake Bay from SAR images. After refinement, the detected shoreline pixels matched those of the original image to within a pixel or two. Mason and Davenport [12] developed a semiautomatic coarse-fine resolution processing method for detecting the muddy shorelines in Western Wash from SAR scenes. The detected shoreline positions agree with the ground measurements to within a pixel or so. Niedermeier *et al.* [27] used wavelet and active contour methods to detect the muddy shorelines from SAR images in the Elbe estuary. The error is estimated by comparing the shorelines with a model solution and gives a mean offset of 2.5 pixels. Liu and Jezek [13] integrated the Canny edge detector and locally adaptive thresholding methods to extract the muddy shorelines in the Galveston Bay of the Gulf of Mexico from Landsat ETM+ images. The relative accuracy of the resulting shorelines is within one pixel compared with the human visual interpretation. Ding *et al.* [14] analyzed the performance of two shoreline extraction approaches using SAR data in the intertidal flat in Shanghai and found that the extraction accuracy decreases when there is water within the intertidal flat. Liu *et al.* [28] mixed the region-based and edge-based active contour models to detect the muddy shorelines in the North of San Francisco Bay, and the average offset is larger than 2 pixels. Hong *et al.* [29] presented an automatic sub-pixel shoreline extraction method from EO-1 Hyperion data on the muddy coast of the East China Sea and achieved an accuracy of 0.40 pixels. There are numerous approaches applied to other coast types (see [30]–[34]), some of which reach accuracies within 5 m of mid-resolution satellite imagery.

III. Experimental Materials

A. Study Area and Shoreline Indicator

Shaped by the sediment supply from the Yellow River and ocean dynamics, the Yellow River Delta (YRD) coast (see Fig. 1) has become one of the most active regions experiencing rapid spatiotemporal changes among the river deltas in the world [3]. The total length of the shoreline in the study area is about 350 km. Except for the artificial coastal section (whose length is about 20% of the total length), the rest of the YRD coast is the muddy coast where the beach slopes are extremely gentle, generally from 1/8000 to 1/12 000. The delta coast contains the diurnal and semi-diurnal tidal zones and the tidal dynamics along the coast are complicated [35]. The tidal dynamics along the northern and southern parts of the YRD coast are generally opposite. During image acquisition, the coastal area can experience both flood and ebb tides.

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The sediments of the tidal flats are dominated by mud or a mixture of mud and fine sand. During ebb tide, these fine-grained sediments may retain the remnant water on the newly exposed tidal flats for several hours, leading to extremely low reflectance contrasts between land and sea. The land cover types along the YRD coast include soil, vegetation, artificial bank, and reservoirs and ponds, which also complicate the land–sea contrasts. The waterline (instantaneous shoreline) is selected as the shoreline indicator and the terms "shoreline" and "waterline" are used interchangeably in this article.

B. Data Description

Due to high temporal variations and the problem of accessibility in muddy coasts, acquiring ground validation data at the time of satellite overpass is difficult. Therefore, we choose MODIS Terra data for experiment and Landsat 7 data for validation, because of their closely matched acquisition times. The presented approach can be used for other spatial resolution data, because it does not depend on the characteristics of MODIS data. The MOD02 Swath image (Collection 6, L1B) acquired at 02:55 (UTC) on March 25, 2015 is obtained from the U.S. NASA Goddard Space Flight Center (GSFC). The Landsat 7 ETM+ image (Tier 1) acquired at 02:41 (UTC) on the same day is obtained from the United States Geological Survey (USGS). The pixels of the MODIS image over the study area have very small view zenith angles (sensor zenith: 0.03°–7.01°), close to the nadir-view of the ETM+ image. The difference between the acquisition times of the MODIS and ETM+ images is also very small (14 min). For spatial accuracy, the 30-m pixel resolution of Landsat data is sufficiently high compared with the 250–1000-m pixel resolutions of MODIS data. Hence, we regard the shoreline positions manually interpreted from the ETM+ image as the estimated ground truth (EGT). There are some data gaps caused by the scan-line-corrector (SLC) failure on the ETM+ image; however, most of the gaps have little negative effect on the shoreline interpretation process. In addition, more image pairs are collected to validate the robustness of the work (see the Supplementary Material).

C. Data Preparation

Raw digital numbers (DNs) of the MODIS image are firstly converted to top of atmosphere (TOA) reflectance values. The TOA image is georeferenced to the UTM projection, using the MODIS Reprojection Tool Swath with the nearest-neighbor resampling method. The image sub-setting is performed to produce the sub-image, which is free of clouds in the study area. For the variations in the position between the MODIS and Landsat images, we perform a sub-pixel georeferencing using the single-step discrete Fourier transformation (DFT) algorithm [36], at a precision of 0.01 pixels. The reflectance difference between MODIS band 2 and band 1 is selected to perform the analyses, because of its superior performance in identifying water from land [37], [38].

IV. Proposed Method

A. Shoreline Presentation

In this article, each pixel denotes one class (sea or land). The points between pixels represent the potential shoreline locations, called candidate shoreline points (CSPs), in which the points indicating true shoreline locations are called shoreline points (SPs). Each CSP is surrounded by a small neighborhood. Specifically, for each set of four nearby pixels that form a square, the central point (i.e., CSP) is used to represent the set (see Fig. 2). The shoreline can be defined by a set of SPs selected from the CSPs. The number of SPs can be random, so that they can represent arbitrarily shaped shorelines. In the following steps, we use the CSPs to synthesize information from their 2×2 neighborhoods. The interval between the CSPs is the same as the pixel size.

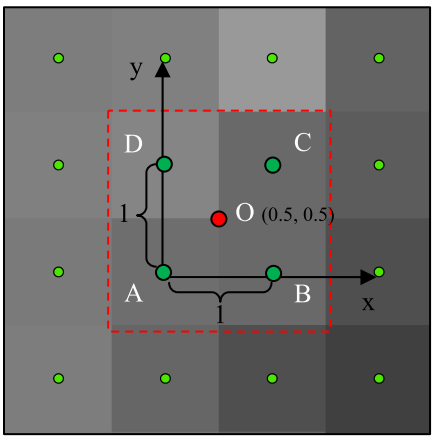


Fig. 2. Diagram of shoreline representation. The background is a section of pixel matrix of the image where each pixel is denoted by its centroid point (small green dot). The large red dot O is the central point (i.e., CSP) of the square (red dashed line) formed by the nearby pixels A–D (large green dots). The numbers 1 and 0.5 indicate the 1- and 0.5-pixel units, respectively. The letters x and y indicate the horizontal and vertical axes of the plane of the pixel matrix, respectively.

B. Shoreline Extraction

Firstly, we evaluate the local possibilities of being boundary, water/land, and shoreline for each CSP, by a set of developed image and spatial indicators. Then, the overall shoreline possibilities are evaluated by an improved min-cost method.

1) Detecting Boundary Strength: We start with the differential analysis (gradient magnitude) because it is the standard means of edge detection. The pixel matrix of reflectance difference values (called pixel values, hereafter) over the study area is assumed as a continuous surface. For each CSP, the first partial derivatives on the x- and y-axes are calculated and the gradient magnitude is determined by the following equation:

$$\text{Gradient} = \sqrt{\left[\frac{\partial f(x,y)}{\partial x}\right]^2 + \left[\frac{\partial f(x,y)}{\partial y}\right]^2} \tag{1}$$

where $f(x, y)$ is a bilinear function

$$f(x,y) = P_A(1-x)(1-y) + P_B x(1-y) + P_C xy + P_D(1-x)y \tag{2}$$

where P_A, P_B, P_C, and P_D are the four nearby pixel values of a specific square, and the gradient is obtained at the central point O (see Fig. 2). When these pixel values are similar, the gradient will be gentle. When the pixel values are different enough,

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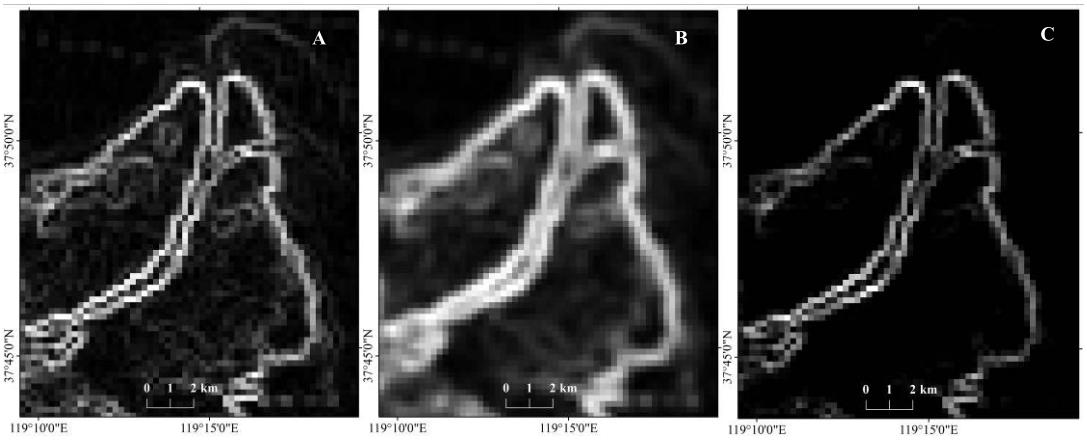


Fig. 3. Illustrations for the image features of (a) gradient, (b) intensity ($\overline{|\mathrm{EI}|}$), and (c) SV. For viewing convenience, the matrix of CSPs is converted to raster maps with the pixel resolution.

the surface will exhibit some high gradient. The reason for choosing the bilinear function is that it is simple in form, continuous, and has simple derivatives [39].

The differential method usually suffers from image noise and does not distinguish textures from boundaries [see Fig. 3(a)]. This limitation can be compensated by local statistical analysis [40]. Therefore, we apply a statistical approach proposed by Pfeifer *et al.* [41], characterized by measuring the heterogeneity of each pixel within its neighborhood. The $3\times$ 3 pixel window is selected as the neighborhood size. Edge intensity (EI) is the maximum value between local and pixel heterogeneity for each pixel and has the sign of pixel heterogeneity

$$\mathrm{EI} = \max(\sigma_N, |\bar{P_N} - P_i|) \times \mathrm{sign}(\bar{P_N} - P_i) \tag{3}$$

where P_i is the value of pixel i, scaled between 0 and 100; σ_N is the standard deviation of the pixel values within the neighborhood N of pixel i; $\bar{P_N}$ is the average of the pixel values within the neighborhood; and $|\bar{P_N} - P_i|$ is the absolute deviation of pixel i from the average. The sign of EI is determined by the sign of pixel deviation $(\bar{P_N} - P_i)$. Pixels with negative EI values have probabilities of being land and pixels with positive EI values belong to water more likely. Absolute EI values range from 0 (no edge intensity) to 100 (maximum edge intensity). Pixels with high absolute EI values are considered to be closer to shoreline positions.

For a given image, boundaries generated from one measure often coincide with boundaries derived from other measures [11]. When discontinuities in EI are correlated with discontinuities in Gradient, the evidence for a boundary increases. In order to integrate EI with Gradient, we use the same set of four nearby pixels that form a square. The average of the absolute EI values of these pixels indicates the representative edge intensity for the square

$$\overline{|\mathrm{EI}|} = \left(\sum_{i=1}^{4} |\mathrm{EI}_i|\right)/4. \tag{4}$$

Because $\overline{|\mathrm{EI}|}$ pools information over the larger spatial support and the pixels around boundaries all have large intensity values, the smooth and spatially extended output is produced [see Fig. 3(b)].

We combine Gradient and $\overline{|\mathrm{EI}|}$ of each square into a hybrid map via point-wise multiplication

$$\mathrm{SV} = \mathrm{Gradient} \times \overline{|\mathrm{EI}|} \tag{5}$$

where salient value (SV) implies the possibility of being a high-confidence edge point. The multiplication operation enlarges the difference between points with strong boundary strength and those with relatively weak strength, and in the meantime suppresses image noise and texture edges [see Fig. 3(c)].

2) Measuring the Possibility of Being Water/Land: One of the fundamental shoreline characteristics is that it separates land and sea regions. Because a CSP represents a local neighborhood, we evaluate the regional difference (RD) within the neighborhood, using the sign of EI values. In order to separate the magnitude of edge intensity from the land/water class, we normalize the EI values of the four pixels within each CSP neighborhood to have the same mean, using a technique called divisive normalization [42]. Then, we sum the normalized EI values as follows:

$$\mathrm{RD} = \sum_{i=1}^{4} (\mathrm{EI}_i / \overline{|\mathrm{EI}|}). \tag{6}$$

The range of RD is between -4 and $+4$. Because negative EI values indicate land possibilities and positive ones water possibilities, the RD values close to 0 are more likely representative of shoreline possibilities, whereas larger negative (positive) values indicate higher land (water) possibilities (see Fig. 4).

3) Spatially Distinguishing Shoreline From Other Spectrally Similar Boundaries: As illustrated by the land cover maps of the YRD coast [43], aquaculture areas, water reservoirs, and brine ponds are widely distributed along the coast. The boundaries of these land cover types are also spectrally similar to the shoreline. In order to distinguish the shoreline from them, we firstly define a shoreline zone [see Fig. 5(a)] to narrow the detection coverage. A region-growing algorithm starting from the ocean is applied to the raster SV map, and

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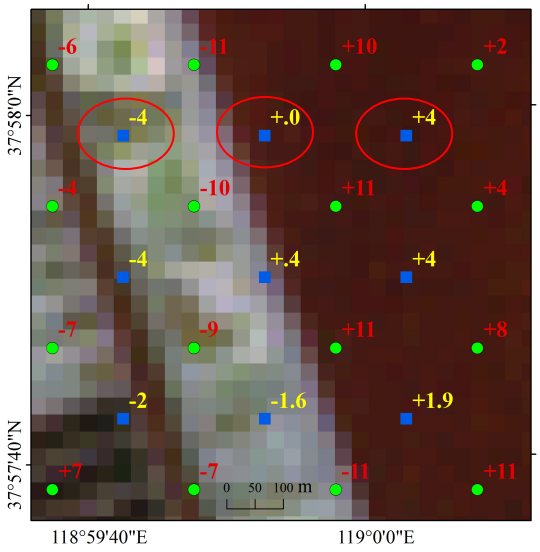


Fig. 4. Examples for the RD indicator. The background is the ETM+ RGB composite image used for reference. Each pixel with its EI value is denoted by a green circle dot. Each CSP with its RD value is denoted by a blue square dot. For example, the RD values of the CSPs within the red circles (from left to right) denote the land, shoreline and water, respectively.

an initial baseline of the shoreline location is produced. The shoreline zone is created from a buffer of the baseline with a 6-pixel width. Then, we introduce a spatial metric called Seaward Distance (SD). In general, the shoreline should be as close as possible to the sea. Hence, the seaward side of the shoreline zone is extracted as the seaward baseline and the nearest distances (NDs) from each CSP to the seaward baseline are calculated as the SDs [see Fig. 5(b)].

4) Synthetic Evaluation of Local Shoreline Possibilities: The above indicators capture different aspects of local shoreline possibilities, but there are some locations where those indicators may conflict with each other. Hence, we make a tradeoff between them by a cost function

$$\text{Cost} = \frac{|\text{RD}|*\text{SD}^m}{(\text{SV}+a)^n} \tag{7}$$

where the SV, RD, and SD values of each CSP are normalized from 0 to 100; a is a constant to avoid the situation where the denominator is zero; higher SVs relate to lower costs, and |RD| and SD values go with costs. The power "n" shows the SV effect; raising the power "n" introduces a very strong inhibition against going through a CSP with relatively lower SV. The power "m" denotes the SD effect, which enhances the boundaries close to the sea whereas suppresses those far away from the sea. If two CSPs have similar boundary strength, the point nearer to the sea will be more likely classified as SP.

For coastal environments where the reflectance contrasts between the land and the sea are large, the reflectance variations in the ocean are relatively small and the shorelines are distinct; raising the power "m" or reducing the power "n" is helpful in capturing those SPs with weak image features. For coastal environments where the land–sea reflectance contrasts are much lower, the shorelines are blurring and the reflectance variations in the sea might be detected as the SPs; raising the power "n" or reducing the power "m" is helpful in avoiding this kind of error.

5) Evaluating Overall Shoreline Possibilities: Based on the cost function values of each CSP, we evaluate the collective properties of the CSPs by searching for min-cost paths. Different settings of parameters m and n lead to different shoreline paths. Thus, an over-complete set of candidate shoreline paths is generated. The set of shoreline paths is found by the uniform cost search (UCS) algorithm [44], an improved version of the Dijkstra algorithm [45]. The cost of a shoreline path $P_i(\text{SP}_s,\text{SP}_e)$ from the starting point SP_s to the endpoint SP_e is calculated as follows:

$$P_i\ (\text{SP}_s, \text{SP}_e) = \sum_{j=s}^{e} \text{Cost}(\text{SP}_j), \quad \text{SP}_j \in P_i. \tag{8}$$

The starting point SP_s and the endpoint SP_e are selected from the CSPs with high SVs. Inspired by Rodriguez and Laio [46], we assume that these CSPs are surrounded by neighbors with lower SVs and they are at a relatively large distance from any other CSPs with higher SVs. For CSP i, we compute its minimum distance δ_i from any other CSP j with a higher SV S_j

$$\delta_i = \min(d_{ij}), \quad j : S_j > S_i. \tag{9}$$

For CSP i with the highest SV, we stipulate its $\delta_i = \max(\delta_j)$, $j \neq i$. Only CSPs with local or global maxima in SV have δ values that are much larger than the usual nearest neighbor distances. $\gamma_i = S_i * \delta_i$ is calculated and sorted in a descending order. The CSPs with anomalously large γ values are used as candidate SP_s and SP_e (e.g., the first-level points in Fig. 6). In order to cover the whole study area, we use more CSPs with high γ values (e.g., the second-level points in Fig. 6). These points are validated by the RD indicator to ensure they lie along land–water boundaries.

C. Parameter Analyses

1) Frequency of Occurrence: Firstly, N min-cost path results $P_j (j = 1, \ldots, N)$ are obtained from N combinations of the m and n parameters, in which SPs are defined as "1" (P_{j1}) and the rest as "0" (P_{j0}). The sampling range of parameters should be sufficiently large to contain a wide variety of path results from noisy to sparse ones [47]. Here, we implement 100 m and n combinations ($1 <= m, n <= 10$), which seem to reasonably contain such a range. A CSP identified as SP in all paths has the highest occurrence value N, whereas a CSP identified as SP in none path has the lowest value 0. A histogram of frequency of occurrence for $N = 100$ is presented in Fig. 7(a). From a statistical perspective, CSPs with higher occurrence values are considered to be more related to SPs.

2) Overall Correspondence: A possible occurrence value i ($0 \leq i <$ maximum of occurrence) can be used as a threshold to produce a potential ground truth (PGT_i), in which the CSPs with the occurrence values higher than i are SPs and defined as "1" (PGT_{i1}) and the rest as "0" (PGT_{i0}). Each PGT_i is compared with each path P_j and generates three outcomes:$\text{TP}_{\text{PGT}_i, P_j}$ (true positive), $\text{FP}_{\text{PGT}_i, P_j}$ (false positive), and $\text{FN}_{\text{PGT}_i, P_j}$ (false negative). Then, these possibilities are

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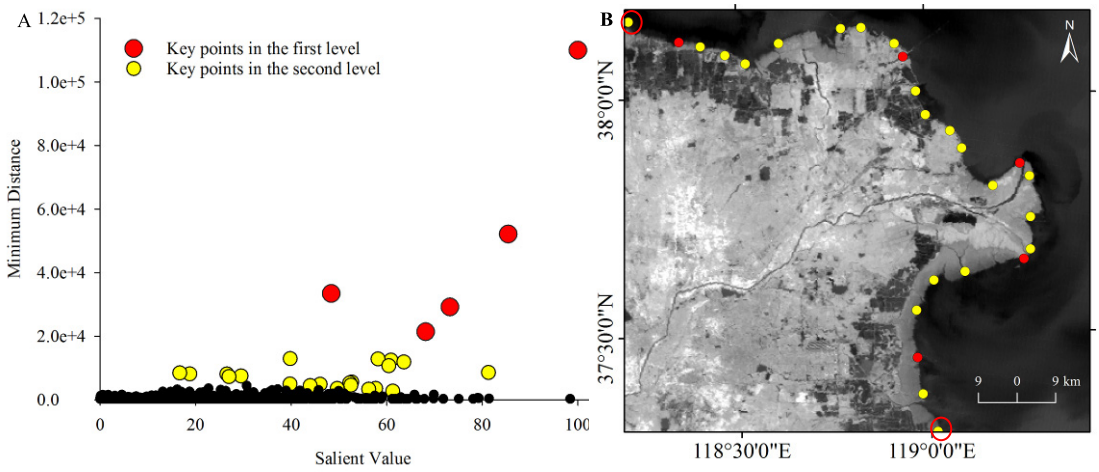


Fig. 5. Shoreline zone and the SD indicator. The background is the raster SV image. (a) Shoreline zone of the study area is surrounded by the red and green contours, in which the green line is the seaward baseline. (b) Detailed view of a local part of the shoreline zone. The blue points are the CSPs and the distance (the red straight line) between one of the CSPs and its nearest point on the seaward baseline (the yellow dots) is the SD.

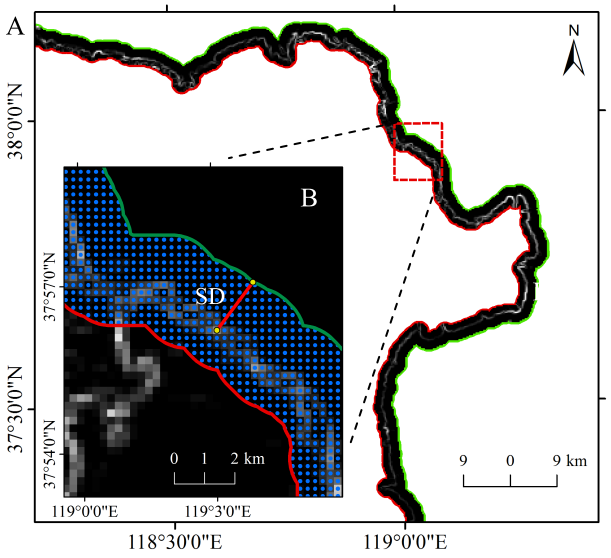


Fig. 6. Illustrations for selecting SP_s and SP_e. (a) Minimum distance (δ) as a function of SV for each CSP. The CSPs with anomalously large δ and SV are classified as the key points in the first level. The points in the second level are also included to cover the entire study area. (b) Geographical distribution of the key points. The points in the red circles are selected as SP_s and SP_e in this article.

averaged over N paths to produce $\overline{\mathrm{TP}}_{\mathrm{PGT}_i}$, $\overline{\mathrm{FP}}_{\mathrm{PGT}_i}$, and $\overline{\mathrm{FN}}_{\mathrm{PGT}_i}$

$$\overline{\mathrm{TP}}_{\mathrm{PGT}_i} = \frac{1}{N}\sum_{j=1}^{N}\mathrm{TP}_{\mathrm{PGT}_i,P_j} = \frac{1}{N}\sum_{j=1}^{N}\left(\frac{1}{\mathrm{Num_CSP}}\sum_{1}^{\mathrm{Num_CSP}}\mathrm{PGT}_{i1}\bigcap P_{j1}\right) \quad (10)$$

$$\overline{\mathrm{FP}}_{\mathrm{PGT}_i} = \frac{1}{N}\sum_{j=1}^{N}\mathrm{FP}_{\mathrm{PGT}_i,P_j} = \frac{1}{N}\sum_{j=1}^{N}\left(\frac{1}{\mathrm{Num_CSP}}\sum_{1}^{\mathrm{Num_CSP}}\mathrm{PGT}_{i1}\bigcap P_{j0}\right) \quad (11)$$

$$\overline{\mathrm{FN}}_{\mathrm{PGT}_i} = \frac{1}{N}\sum_{j=1}^{N}\mathrm{FN}_{\mathrm{PGT}_i,P_j} = \frac{1}{N}\sum_{j=1}^{N}\left(\frac{1}{\mathrm{Num_CSP}}\sum_{1}^{\mathrm{Num_CSP}}\mathrm{PGT}_{i0}\bigcap P_{j1}\right) \quad (12)$$

where Num_CSP is the total number of CSPs within the shoreline zone. If $\overline{\mathrm{TP}}_{\mathrm{PGT}_i}$ is high and $\overline{\mathrm{FP}}_{\mathrm{PGT}_i}$ and $\overline{\mathrm{FN}}_{\mathrm{PGT}_i}$ are low, the overall correspondence between PGT_i and the path set is high. A Chi-square test can measure this correspondence [48]

$$\chi^2_{\mathrm{PGT}_i} = \frac{(\mathrm{TPR}_{\mathrm{PGT}_i} - Q_{\mathrm{PGT}_i})}{(1-Q_{\mathrm{PGT}_i})} * \frac{(Q_{\mathrm{PGT}_i} - \mathrm{FPR}_{\mathrm{PGT}_i})}{Q_{\mathrm{PGT}_i}} \quad (13)$$

where $Q_{\mathrm{PGT}_i} = \overline{\mathrm{TP}}_{\mathrm{PGT}_i} + \overline{\mathrm{FP}}_{\mathrm{PGT}_i}$, $P_{\mathrm{PGT}_i} = \overline{\mathrm{TP}}_{\mathrm{PGT}_i} + \overline{\mathrm{FN}}_{\mathrm{PGT}_i}$, $\mathrm{TPR}_{\mathrm{PGT}_i} = (\overline{\mathrm{TP}}_{\mathrm{PGT}_i})/(P_{\mathrm{PGT}_i})$, and $\mathrm{FPR}_{\mathrm{PGT}_i} = (\overline{\mathrm{FP}}_{\mathrm{PGT}_i})/(1 - P_{\mathrm{PGT}_i})$. The χ^2 measure gives the overall correspondence between the frequency of occurrence and the shoreline path set [see Fig. 7(b)].

3) Parameter Sensitivity: In this section, we firstly investigate which combination of m and n parameters gives the best match to the PGTs with high χ^2 values, and the parameter sensitivity to the shoreline paths can be evaluated. The matches between each path P_j and the PGT_i are also measured by Chi-square tests with $\mathrm{TP}_{P_j,\mathrm{PGT}_i}$, $\mathrm{FP}_{P_j,\mathrm{PGT}_i}$, and $\mathrm{FN}_{P_j,\mathrm{PGT}_i}$

$$\chi^2_{P_j,\mathrm{PGT}_i} = \frac{(\mathrm{TPR}_{P_j,\mathrm{PGT}_i} - Q_{P_j,\mathrm{PGT}_i})}{(1-Q_{P_j,\mathrm{PGT}_i})} * \frac{(Q_{P_j,\mathrm{PGT}_i} - \mathrm{FPR}_{P_j,\mathrm{PGT}_i})}{Q_{P_j,\mathrm{PGT}_i}} \quad (14)$$

where $Q_{P_j,\mathrm{PGT}_i} = \mathrm{TP}_{P_j,\mathrm{PGT}_i} + \mathrm{FP}_{P_j,\mathrm{PGT}_i}$, $P_{P_j,\mathrm{PGT}_i} = \mathrm{TP}_{P_j,\mathrm{PGT}_i} + \mathrm{FN}_{P_j,\mathrm{PGT}_i}$, $\mathrm{TPR}_{P_j,\mathrm{PGT}_i} = (\mathrm{TP}_{P_j,\mathrm{PGT}_i}/P_{P_j,\mathrm{PGT}_i})$, and $\mathrm{FPR}_{P_j,\mathrm{PGT}_i} = (\mathrm{FP}_{P_j,\mathrm{PGT}_i}/1 - P_{P_j,\mathrm{PGT}_i})$. For each PGT, the parameter setting with the highest χ^2 value is obtained and the shoreline path produced by these parameters is the best match to the PGT. Finally, we evaluate the sensitivity of parameter settings to different detection results.

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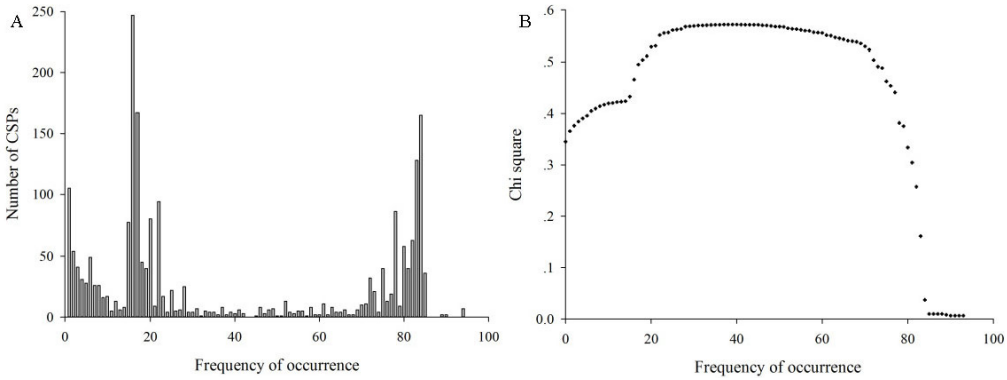


Fig. 7. Parameter analyses. (a) Distribution of frequency of occurrence of CSPs obtained from the sum of 100 shoreline paths produced by 100 combinations of m and n parameters. (b) The Chi square test measures the overall correspondence between each PGT and the shoreline path set.

D. Evaluation Strategies

1) Comparing Detected Shoreline Versus EGT: At each SP on a shoreline path, the distance to the closest point (CP) on the reference shoreline $d(\mathrm{SP}, \mathrm{CP})$ is considered as an indication of the spatial accuracy of the shoreline at that SP, called ND. For a specific distance x, we compute the proportion p of the SPs, whose $d(\mathrm{SP}, \mathrm{CP})$ is less than x (see Fig. 8). Mason *et al.* [49] have shown that the extraction accuracy within one pixel is sufficient for practical applications. Hence, we set one-pixel unit as the benchmark.

2) Comparing the Presented Approach Versus Other Methods: Among the current state-of-the-art computational approaches of edge detection, the Canny edge detector is widely adopted for building reference images, because there is a consensus about its superior performance. Thus, we choose the Canny method for a quantitative comparison. The Canny detection process is implemented in MATLAB 2015b software. The detection result is a binary edge map. The edge pixels within the shoreline zone are converted to the edge points and used for the comparison. For the methods designed specifically for the shoreline extraction purpose, there are no off-the-shelf tools to perform a quantitative comparison at the current stage. Hence, we make a discussion with related works based on the literature review.

3) Validating the Effectiveness of the Presented Approach for Different Data and Environments: Another Landsat 5 TM image obtained on October 7, 2008 is used to check the ability of the presented approach for different spatial resolution data and coastal environments. In this image, the coastal environment that lies to the North of the Yellow River Estuary (YRE, Fig. 1) shows relatively large land–sea reflectance contrasts, whereas the coastal environment that lies to the South of the YRE shows much lower land–sea contrasts. The presented approach and human interpretation are both applied to the image to extract the shorelines. The comparison between the automatic detection and human-derived results is made in the same way as above.

V. Results and Discussion

A. Parameter Analyses

One-hundred shoreline paths are produced using 100 combinations of m and n parameters. There are 11363 CSPs

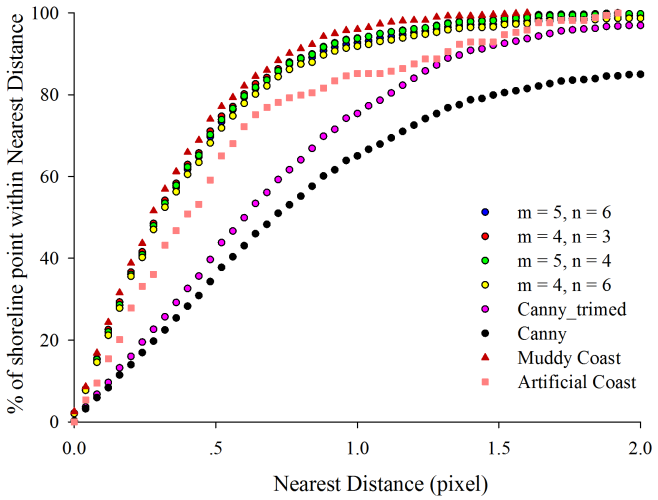


Fig. 8. Plots of the percentages of SPs within the NDs less than 2.0 pixel for selected m and n parameter settings, the Canny detection results, and the artificial and muddy coasts ($m = 4$, $n = 3$).

(excluding the starting and end points) in the shoreline zone and 9184 points (80.8%) of them have zero occurrences. Fig. 7(a) shows the distribution of frequency of occurrence of each CSP as SP, in which two clusters of SPs appear. One cluster is approximately located in the occurrence range 70–85 and the other below 25. This bimodal pattern clearly reveals the huge difference between the SPs with high occurrence values (more related to true SPs) and those with low values (more related to false SPs). In addition, there are a few SPs within the range 25–70, indicating that their shoreline possibilities are not high enough. This statistical result provides a basis for a distinction between true and false SPs. Fig. 7(b) shows the Chi-square tests for each PGT. The best PGT_i occurs at the occurrence value $i = 39$ that has the highest χ^2 value. However, in the range 22–71, the χ^2 values are not significantly different (mean = 0.56, standard deviation = 0.01). The low χ^2 values in the range below 22 and above 71 are mainly due to their high FPs and FNs, respectively. Table I shows the χ^2 values for the best parameter settings matched to the $\mathrm{PGT}_i (22 \leq i \leq 71)$. The first 33 highest χ^2 values are all produced by the parameters ($m = 5$, $n = 6$), which are the best parameter settings for the PGTs in the range

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TABLE I

CHI-SQUARE MEASUREMENTS FOR THE BEST PARAMETER SETTINGS FOR $PGT_I (22 \leq i \leq 71)$

PGT_i	m	n	Chi-square	PGT_i	m	n	Chi-square
41	5	6	0.988818434	**53**	5	6	0.933200237
42	5	6	0.987577856	**54**	5	6	0.929509187
43	5	6	0.987577856	**55**	5	6	0.923362101
44	5	6	0.987577856	**56**	5	6	0.917220834
45	5	6	0.986336942	**27**	4	6	0.916155709
40	5	6	0.98142664	**57**	5	6	0.915993279
39	5	6	0.980226803	**26**	4	6	0.909837952
46	5	6	0.976418131	**58**	5	6	0.906181196
38	5	6	0.975377451	**25**	5	7	0.905760205
37	5	6	0.972969326	**59**	5	6	0.903730496
47	5	6	0.972702468	**60**	5	6	0.901280723
36	5	6	0.970721063	**61**	5	6	0.887823518
35	5	6	0.968350492	**24**	4	7	0.886765076
48	5	6	0.965277497	**62**	5	6	0.885379758
34	5	6	0.963641206	**23**	7	6	0.882825707
33	5	6	0.9613781	**63**	4	4	0.876703932
32	5	6	0.957982715	**64**	4	4	0.871820482
31	5	6	0.956829911	**22**	3	6	0.871680146
49	5	6	0.956625727	**65**	4	4	0.866940716
50	5	6	0.955390699	**66**	4	4	0.859627967
51	5	6	0.954155906	**67**	4	4	0.857192222
30	4	6	0.95239256	**68**	4	4	0.854757395
29	4	6	0.947860851	**69**	4	4	0.847458416
28	4	6	0.943368634	**70**	4	4	0.835311764
52	5	6	0.938124903	**71**	4	4	0.821976817

31–62, and ($m = 4$, $n = 6$) which are the best ones for the PGTs in the range 26–30. This phenomenon illustrates that the parameter settings are relatively stable with respect to changes in shoreline paths under normal circumstances.

The above statistical result is based on the accordance of several candidate paths as a reference and measures the similarity between a candidate path and the entire path set. The two selected parameter settings are the optimal matches of this collective property. However, as Lopez-Molina *et al.* [50] pointed out, this method penalizes those candidate paths with superior performance which detect SPs that most of the other paths cannot find. By fine-tuning the optimal correspondence parameter settings, a more accurate result can be achieved. For the same $n = 6$, the χ^2 value for $m = 5$ is higher than that for $m = 4$, indicating a more important SD effect. There are large land–sea reflectance contrasts and small reflectance variations in the sea. Hence, we suppress the SV effect and enhance the SD effect by reducing the power "n" below the power "m." Specifically, the above two parameter settings are adjusted to ($m = 5, n = 4$) and ($m = 4, n = 3$), respectively. The newly adjusted parameter settings are hoped to screen for SPs with relatively weak image features. The reason why we reduce "n" rather than increase "m" is that raising the powers will further weaken the importance of the RD indicator. For the same reason, we prefer ($m = 4, n = 3$) between the adjusted parameter settings, in which the relatively low powers of "m" and "n" tend to balance the importance of each indicator.

TABLE II

PERCENTAGES OF SPS WITHIN TYPICAL NDS FOR SELECTED PARAMETER SETTINGS AND THE CANNY ALGORITHM

ND (pixel)	% of SPs within ND						
	m=4, n=3	m=5, n=4	m=5, n=6	m=4, n=6	Canny trimmed	Artificial	Muddy
0.20	36.60	36.01	36.33	35.58	15.98	27.81	38.79
0.40	62.93	62.34	61.78	60.52	32.59	50.89	65.93
0.60	80.17	79.69	79.41	77.90	49.91	72.19	82.15
0.80	89.02	88.90	88.64	87.47	64.09	79.88	91.30
1.00	93.86	93.74	92.66	91.84	75.40	85.21	96.02
1.20	96.10	96.10	95.15	94.44	84.02	87.57	98.23
1.40	97.99	97.87	97.04	96.45	90.84	92.90	99.26
1.60	99.17	98.82	97.99	97.40	93.72	95.86	100.00
1.80	99.65	99.41	98.70	98.23	96.05	98.22	
1.92	100.00	99.76	99.29	98.70	96.77	100.00	
2.00		99.76	99.29	98.70	96.95		
2.40		100.00	99.65	99.17	97.94		
3.28			100.00	99.88	99.10		
3.56				100.00	99.19		
4.80					100.00		

B. Accuracy Assessment

Fig. 8 and Table II show the percentages of SPs within the NDs for the above-selected parameter settings. The fine-tuned parameters ($m = 4, n = 3$) and ($m = 5, n = 4$) have similar accuracies in all NDs. More than 93% of their SPs are located within 1.0 pixel (250 m). The mean (standard deviation) distances from the SPs for the parameters ($m = 4, n = 3$) and ($m = 5, n = 4$) to the EGT are 0.38 ($\pm$0.34) pixels (95.70 ($\pm$84.86) m) and 0.39 ($\pm$0.35) pixels (97.48 ($\pm$87.33) m), respectively. The biggest difference between the shoreline paths produced by them is located in the estuary area [see Fig. 9(a)], where the RD indicator penalizes the CSPs with relatively high SVs but is dominated by high possibilities of being land or water. The lower powers of "m" and "n" emphasize the importance of RD, and thereby increase the sinuosity of the shoreline. This characteristic is different from ordinary edge detection methods that usually impose a constraint of smoothness due to the ill-posed nature of the methods.

The optimal parameter settings for the overall correspondence ($m = 5$, $n = 6$) and ($m = 4$, $n = 6$) also have high accuracies. More than 92% and 91% of their SPs are located within 1.0 pixel. The mean (standard deviation) distances from the SPs for the parameters ($m = 5$, $n = 6$) and ($m = 4$, $n = 6$) to the EGT are 0.40 ($\pm$0.39) pixels (100.83 ($\pm$97.66) m) and 0.42 ($\pm$0.43) pixels (106.18 ($\pm$107.58) m), respectively. Besides the RD effects illustrated in Fig. 9(a),

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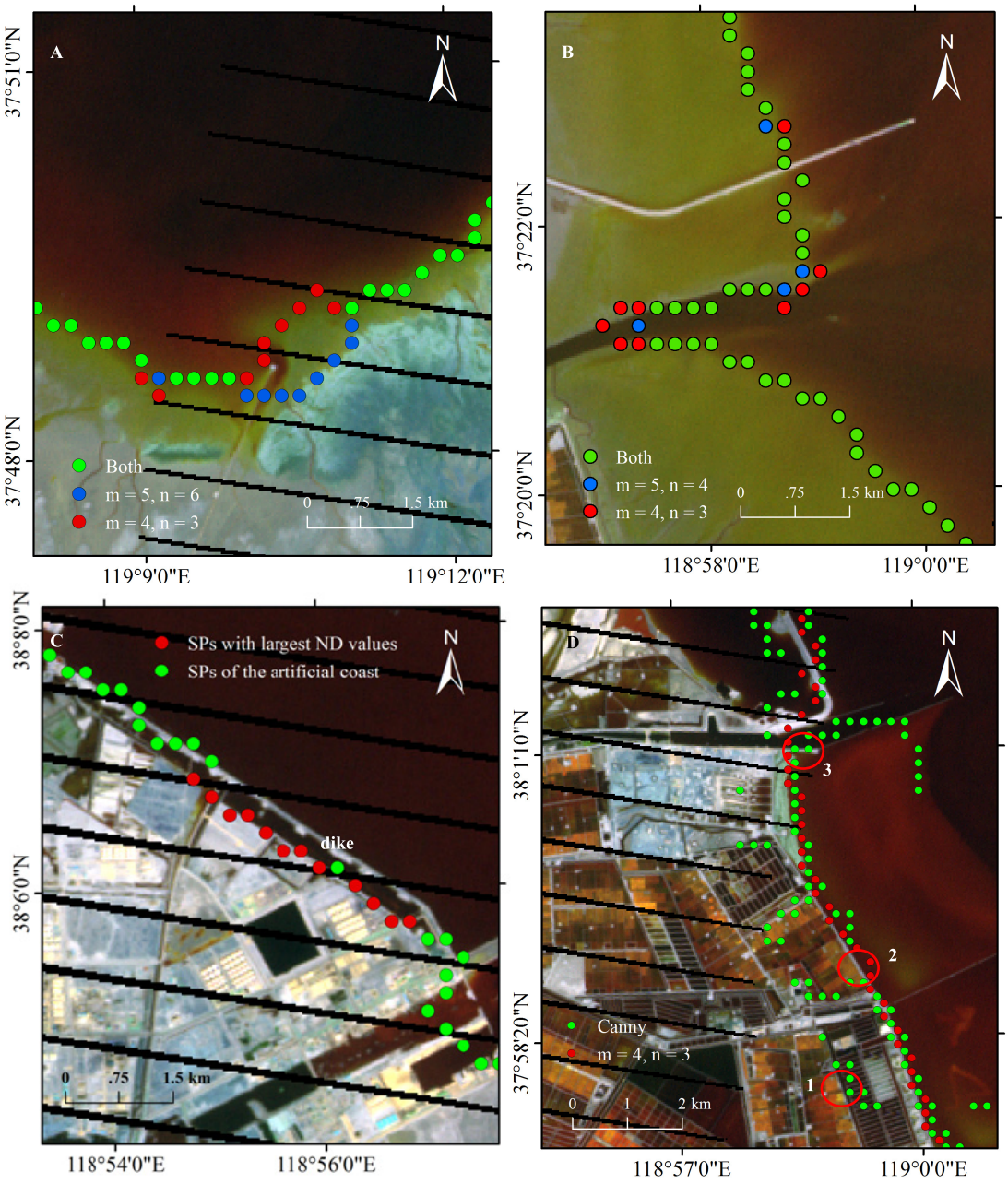


Fig. 9. Shoreline detection results for the MODIS image. The background is the ETM+ RGB composite map. (a) and (b) Red points are the SPs produced only by the parameters ($m = 4$, $n = 3$); the green points are the SPs produced by both parameter settings; and the blue points are the SPs produced only by other parameter settings. (c) Red and green points are the SPs in the artificial coast produced by the parameters ($m = 4$, $n = 3$), in which the red points are SPs with the largest ND values. (d) Green points are the edge points detected by the Canny method and the red points are the SPs produced by the parameters ($m = 4$, $n = 3$). The red circles with numbers 1–3, respectively, indicate the topological errors of insufficient length, discontinuity, and thick edge that occurred in the Canny detection result.

the biggest difference between the shoreline paths produced by the optimally adjusted parameters ($m = 4$, $n = 3$) and the optimal correspondence parameters ($m = 5$, $n = 6$) is the two different sets of SPs in the tidal flat [see Fig. 9(b)]. Because the SV effect is dominant in the parameters ($m = 5$, $n = 6$), the path tends to ignore the front area where the image signals are faint. However, the SD effect is enhanced in the parameters ($m = 4$, $n = 3$) and the CSPs with relatively low image features are selected as SPs since they are closer to the sea.

Fig. 8 and Table II compare the SP accuracies between the artificial and muddy coasts. For the shoreline path produced by the parameters ($m = 4$, $n = 3$), 169 (19.95%) and 678 (80.05%) SPs belong to the artificial and muddy coasts, respectively. 96.02% of the SPs in the muddy coast are located within 1.0 pixel, outperforming the SP accuracies of the artificial coast (85.21%). The mean (standard deviation) distances from the SPs in the muddy and artificial coasts to the EGT are 0.35 (±0.29) pixels (87.25 (±73.72) m) and 0.52 (±0.45) pixels (129.62 (±113.36) m), respectively.

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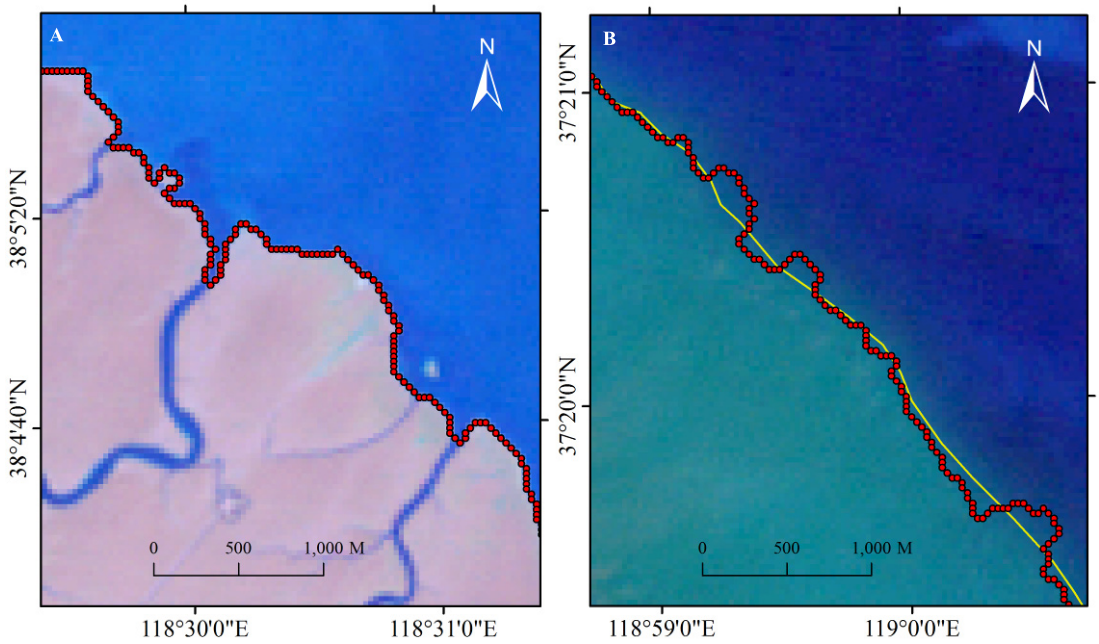


Fig. 10. Shoreline detection results for the Landsat TM image. The red dots are the SPs produced by the proposed approach and the yellow line is the human-derived shoreline. (a) Detected shoreline in the coastal environment with high land–sea reflectance contrast. (b) Detected shorelines in the coastal environment with low land–sea contrast.

We identify the SPs in the artificial coast with the highest ND values and find that most of them are distributed along the land–water boundaries related to the inland [see Fig. 9(c)]. Some narrow and long dikes have been constructed in the sea and finally convert some part of seawater into inland water. Because these dikes are too narrow (the widths are usually less than 0.40 pixels) and relatively far from the inland, the image resolution cannot capture enough signals from these dikes.

C. Comparison With Other Methods

The comparison between the presented approach and the Canny method is performed from qualitative and quantitative perspectives. For the degree of human intervention, although the Canny detection range is confined to the shoreline zone, there is much post-processing work that should be done manually [see Fig. 9(d)], whereas the shoreline paths produced by our approach involves little post-processing work. For the geometric shape, the topological characteristics of the shoreline cannot be completely maintained by the Canny detector. According to Tan *et al.* [51], the line should be continuous, thin, and have sufficient length. The red circles in Fig. 9(d) illustrate the topological errors of insufficient length (number 1), discontinuity (number 2), and thick edge (number 3). Due to the global nature of the min-cost path algorithm, our results are free of these errors. For the positional accuracy, Fig. 8(a) and Table II show the accuracies of the original and manually trimmed edge points produced by the Canny detector. 75.40% of the trimmed edge points are within 1.0 pixel. The mean (standard deviation) distances from the trimmed edge points to the EGT are 0.72 (±0.61) pixels (180.71 (±151.64) m). These accuracies are obviously lower than those produced by our approach.

Based on the literature review of the related works on the pixel-level approaches applied to muddy shoreline detection, it is seen that the positional accuracies of our approach are among the highest level. The deviations are small compared to the amplitude of waterline fluctuations and improvement is achieved for per-pixel accuracies of muddy shoreline extraction. In addition, the accuracy assessment is applied to the shoreline, which is about 350 km in length, much longer than those in most other studies. This spatial coverage can accommodate a variety of coastal environments and provide sufficient experimental materials for testing the generality of the presented approach.

D. Effectiveness of the Presented Approach for Different Data and Environments

We apply the above approach to the Landsat 5 TM image in 2008, which exhibits more complicated coastal environments. The detection result shows that the optimal parameter settings for the overall correspondence are ($m = 4$, $n = 6$) and ($m = 5$, $n = 6$), which are the best parameter settings for the PGTs with the first 23 highest χ^2 values. The mean (standard deviation) distances from the SPs produced by them to the human-derived shoreline are 0.68 (±1.04) pixels (20.33 (±31.22) m) and 0.77 (±1.20) pixels (22.98 (±35.98) m), respectively. As shown in Fig. 10(a), the extracted shoreline is in close agreement with the original image in the coastal environment where the land–sea reflectance contrast is large. The main divergences between the automatic detection and human interpretation results come from the coastal environment lies to the south of the YRE where the land–sea reflectance contrast is extremely low. For the same $n = 6$, the χ^2 value for $m = 4$ is higher than that for $m = 5$, indicating less important SD

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effect. Due to the much lower reflectance contrasts between the tidal flat and seawater, the high SD effect may enhance the possibility that the reflectance variations in the sea are detected as SPs. Therefore, we reduce the SD effect to $m = 1$, and the mean (standard deviation) distances from the produced SPs to the human-derived shoreline are 0.51 ($\pm$0.64) pixels (15.15 ($\pm$19.12) m). The main divergences occur where the human eye has difficulties in determining the shoreline positions [see Fig. 10(b)]. On the whole, the proposed approach has the ability to perform as well as for human interpretation. In addition, the automatic approach is more accurate in locating the shoreline in detail than human interpretation.

VI. Conclusion

This article presents an indicator and min-cost approach to detect the muddy shoreline from satellite images of the YRD coast, China. A new shoreline representation is proposed and a set of image and spatial indicators are developed to assess local shoreline probabilities. A cost function is formulated to combine these indicators and a set of shoreline paths is produced by an improved min-cost path method, to evaluate the overall shoreline possibilities. The optimal shoreline paths are selected based on the parameter analyses of the shoreline set. The average distances from the detected SPs to the ground truth are within 0.5 pixels, and over 90% of the SPs are located within 1.0 pixel. The presented approach is based on analytical computing and uses as few parameters as possible, showing the advantages of simplicity in computation, ease of implementation, little human intervention, and a certain level of generality that can be used for different spatial resolution data and coastal environments. It can serve as the operational workflow for automatic shoreline mapping or input for subpixel shoreline extraction.

References

[1] E. H. Boak and I. L. Turner, "Shoreline definition and detection: A review," *J. Coastal Res.*, vol. 214, pp. 688–703, Jul. 2005.

[2] B. W. Flemming, "Geographic distribution of muddy coasts," in *Muddy Coasts World: Processes, Deposits Function*, T. Healy, Y. Wang, and J.-A. Healy, Eds. Amsterdam, The Netherlands: Elsevier, 2002, pp. 99–202.

[3] Y. Zhang, "Coastal environmental monitoring using remotely sensed data and GIS techniques in the Modern Yellow River delta, China," *Environ. Monit. Assessment*, vol. 179, nos. 1–4, pp. 15–29, Aug. 2011.

[4] Y. Zhang, "Environmental monitoring of spatial-temporal changes using remote sensing and GIS techniques in the Abandoned Yellow River Delta coast, China," *Int. J. Environ. Pollut.*, vol. 45, no. 4, pp. 327–341, 2011.

[5] A. M. Muslim, G. M. Foody, and P. M. Atkinson, "Shoreline mapping from coarse–spatial resolution remote sensing imagery of seberang Takir, Malaysia," *J. Coastal Res.*, vol. 236, pp. 1399–1408, Nov. 2007.

[6] Z. Chu, X. Sun, S. Zhai, and K. Xu, "Changing pattern of accretion/erosion of the modern Yellow River (Huanghe) subaerial delta, China: Based on remote sensing images," *Mar. Geol.*, vol. 227, nos. 1–2, pp. 13–30, Mar. 2006.

[7] L. Feng, C. Hu, X. Chen, X. Cai, L. Tian, and W. Gan, "Assessment of inundation changes of Poyang Lake using MODIS observations between 2000 and 2010," *Remote Sens. Environ.*, vol. 121, pp. 80–92, Jun. 2012.

[8] A. Moradi, L. Metivier, O. De Viron, S. Calmant, and C. Mering, "Evaluation of MODIS data for improved monitoring of the Caspian Sea," *Int. J. Remote Sens.*, vol. 35, no. 16, pp. 6060–6075, Aug. 2014.

[9] A. Seale, P. Christoffersen, R. I. Mugford, and M. O'leary, "Ocean forcing of the Greenland Ice Sheet: Calving fronts and patterns of retreat identified by automatic satellite monitoring of eastern outlet glaciers," *J. Geophys. Res.*, vol. 116, no. F3, Aug. 2011, Art. no. F03013, doi: 10.1029/2010jf001847.

[10] J. F. Canny, "A computational approach to edge detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 8, no. 6, pp. 679–698, Nov. 1986.

[11] M. Leordeanu, R. Sukthankar, and C. Sminchisescu, "Generalized boundaries from multiple image interpretations," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 36, no. 7, pp. 1312–1324, Jul. 2014.

[12] D. Mason and I. Davenport, "Accurate and efficient determination of the shoreline in ERS–1 SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 34, no. 5, pp. 1243–1253, Sep. 1996.

[13] H. Liu and K. C. Jezek, "Automated extraction of coastline from satellite imagery by integrating canny edge detection and locally adaptive thresholding methods," *Int. J. Remote Sens.*, vol. 25, no. 5, pp. 937–958, Mar. 2004.

[14] X. Ding, F. Nunziata, X. Li, and M. Migliaccio, "Performance analysis and validation of waterline extraction approaches using single- and dualpolarimetric SAR data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 3, pp. 1019–1027, Mar. 2015.

[15] T. Pavlidis and Y.-T. Liow, "Integrating region growing and edge detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 12, no. 3, pp. 225–233, Mar. 1990.

[16] Y. Zhang and S. L. Chen, "Super-resolution mapping of coastline with remotely sensed data and geostatistics," *J. Remote Sens.*, vol. 14, no. 1, pp. 148–164, Jan. 2010.

[17] N. J. Murray *et al.*, "The global distribution and trajectory of tidal flats," *Nature*, vol. 565, no. 7738, pp. 222–225, Jan. 2019, doi: 10.1038/s41586-018-0805-8.

[18] S. Seo, "Subpixel edge localization based on adaptive weighting of gradients," *IEEE Trans. Image Process.*, vol. 27, no. 11, pp. 5501–5513, Nov. 2018.

[19] A. J. Tabatabai and O. R. Mitchell, "Edge location to subpixel values in digital imagery," *IEEE Trans. Pattern Anal. Mach. Intell.*, vols. PAMI–6, no. 2, pp. 188–201, Mar. 1984.

[20] E. Lyvers, O. Mitchell, M. Akey, and A. Reeves, "Subpixel measurements using a moment-based edge operator," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 11, no. 12, pp. 1293–1309, Dec. 1989.

[21] M. Hagara and P. Kulla, "Edge detection with sub-pixel accuracy based on approximation of edge with ERF function," *Radio Eng.*, vol. 20, no. 2, pp. 516–524, Jun. 2011.

[22] J. E. Pardo-Pascual, J. Almonacid-Caballer, L. A. Ruiz, and J. Palomar-Vázquez, "Automatic extraction of shorelines from Landsat TM and ETM+ multi-temporal images with subpixel precision," *Remote Sens. Environ.*, vol. 123, pp. 1–11, Aug. 2012.

[23] T. Fuse and T. Ohkura, "Development of shoreline extraction method based on spatial pattern analysis of satellite SAR images," *Remote Sens.*, vol. 10, no. 9, p. 1361, Aug. 2018, doi: 10.3390/rs10091361.

[24] S. Dellepiane, R. De Laurentiis, and F. Giordano, "Coastline extraction from SAR images and a method for the evaluation of the coastline precision," *Pattern Recognit. Lett.*, vol. 25, no. 13, pp. 1461–1470, Oct. 2004.

[25] T. W. Ryan *et al.*, "Extraction of shoreline features by neural nets and image processing," *Photogramm. Eng. Remote Sens.*, vol. 57, no. 7, pp. 947–955, Jul. 1991.

[26] J.-S. Lee and I. Jurkevich, "Coastline detection and tracing in SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 28, no. 4, pp. 662–668, Jul. 1990.

[27] A. Niedermeier, E. Romaneessen, and S. Lehner, "Detection of coastlines in SAR images using wavelet methods," *IEEE Trans. Geosci. Remote Sens.*, vol. 38, no. 5, pp. 2270–2281, 2000.

[28] C. Liu, Y. Xiao, and J. Yang, "A coastline detection method in polarimetric SAR images mixing the region–based and edge–based active contour models," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3735–3747, Jul. 2017.

[29] Z. Hong *et al.*, "Automatic sub-pixel coastline extraction based on spectral mixture analysis using EO–1 Hyperion data," *Front. Earth Sci.*, vol. 13, no. 3, pp. 478–494, Sep. 2019, doi: 10.1007/s11707-018-0702-5.

[30] K. Vos, M. D. Harley, K. D. Splinter, J. A. Simmons, and I. L. Turner, "Sub-annual to multi-decadal shoreline variability from publicly available satellite imagery," *Coastal Eng.*, vol. 150, pp. 160–174, Aug. 2019.

[31] A. Luijendijk, G. Hagenaars, R. Ranasinghe, F. Baart, G. Donchyts, and F. Aarninkhof, "The state of the World's Beaches," *Sci. Rep.*, vol. 8, p. 6641, Apr. 2018.

[32] L. Mentaschi, M. I. Vousdoukas, J.-F. Pekel, E. Voukouvalas, and L. Feyen, "Global long-term observations of coastal erosion and accretion," *Sci. Rep.*, vol. 8, p. 12876. Aug. 2018.

[33] J. E. Pardo-Pascual *et al.*, "Assessing the accuracy of automatically extracted shorelines on microtidal beaches from landsat 7, landsat 8 and sentinel-2 imagery," *Remote Sens.*, vol. 10, no. 2, p. 326, Feb. 2018.

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[34] J. Almonacid-Caballer *et al.*, "Evaluation of annual mean shoreline position deduced from Landsat imagery as a mid-term coastal evolution indicator," *Mar. Geol.*, vol. 372, pp. 79–88, Feb. 2016.
[35] Y. Zhang, S. L. Chen, and G. C. Gu, "Spatial extent and tidal characteristics of the diurnal tidal zone along the Yellow River Delta coast," *Chin. J. Hydrodyn.*, vol. 30, no. 5, pp. 540–548, Sep. 2015.
[36] M. Guizar-Sicairos, S. T. Thurman, and J. R. Fienup, "Efficient subpixel image registration algorithms," *Opt. Lett.*, vol. 33, no. 2, pp. 156–158, Jan. 2008.
[37] D. Sun, Y. Yu, and M. D. Goldberg, "Deriving water fraction and flood maps from MODIS images using a decision tree approach," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 4, no. 4, pp. 814–825, Dec. 2011.
[38] I. Klein, U. Gessner, A. J. Dietz, and C. Kuenzer, "Global WaterPack–A 250 m resolution dataset revealing the daily dynamics of global inland water bodies," *Remote Sens. Environ.*, vol. 198, pp. 345–362, Sep. 2017.
[39] G. Barbujani, N. L. Oden, and R. R. Sokal, "Detecting regions of abrupt change in maps of biological variables," *Syst. Zool.*, vol. 38, no. 4, pp. 376–389, Dec. 1989.
[40] G. Papari and N. Petkov, "Edge and line oriented contour detection: State of the art," *Image Vis. Comput.*, vol. 29, nos. 2–3, pp. 79–103, Feb. 2011.
[41] M. Pfeifer *et al.*, "Creation of forest edges has a global impact on forest vertebrates," *Nature*, vol. 551, no. 7679, pp. 187–191, Nov. 2017.
[42] M. Carandini and D. J. Heeger, "Normalization as a canonical neural computation," *Nature Rev. Neurosci.*, vol. 13, no. 1, pp. 51–62, Jan. 2012.
[43] M. Ottinger, C. Kuenzer, G. Liu, S. Wang, and S. Dech, "Monitoring land cover dynamics in the yellow river delta from 1995 to 2010 based on landsat 5 TM," *Appl. Geography*, vol. 44, pp. 53–68, Oct. 2013.
[44] A. Felner, "Position paper: Dijkstra's algorithm versus uniform cost search or a case against dijkstra's algorithm," in *Proc.4th Annu. Symp. Combinat. Search (SoCS)*, pp. 47–51, Jul. 2011.
[45] E. W. Dijkstra, "A note on two problems in connexion with graphs," *Numer. Math.*, vol. 1, no. 1, pp. 269–271, Dec. 1959.
[46] A. Rodriguez and A. Laio, "Clustering by fast search and find of density peaks," *Science*, vol. 344, no. 6191, pp. 1492–1496, Jun. 2014.
[47] Y. Yitzhaky and E. Peli, "A method for objective edge detection evaluation and detector parameter selection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 25, no. 8, pp. 1027–1033, Aug. 2003.
[48] H. C. Kraemer, *Evaluating Medical Tests: Objective and Quantitative Guidelines*. Thousand Oaks, CA, USA: Sage, 1992.
[49] D. C. Mason, I. J. Davenport, G. J. Robinson, R. A. Flather, and B. S. Mccartney, "Construction of an inter-tidal digital elevation model by the 'Water-Line' Method," *Geophys. Res. Lett.*, vol. 22, no. 23, pp. 3187–3190, Dec. 1995.
[50] C. Lopez-Molina, B. De Baets, and H. Bustince, "Quantitative error measures for edge detection," *Pattern Recognit.*, vol. 46, no. 4, pp. 1125–1139, Apr. 2013.
[51] H. Tan, S. Gelfand, and E. Delp, "A cost minimization approach to edge detection using simulated annealing," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 14, no. 1, pp. 3–18, Jan. 1992.



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